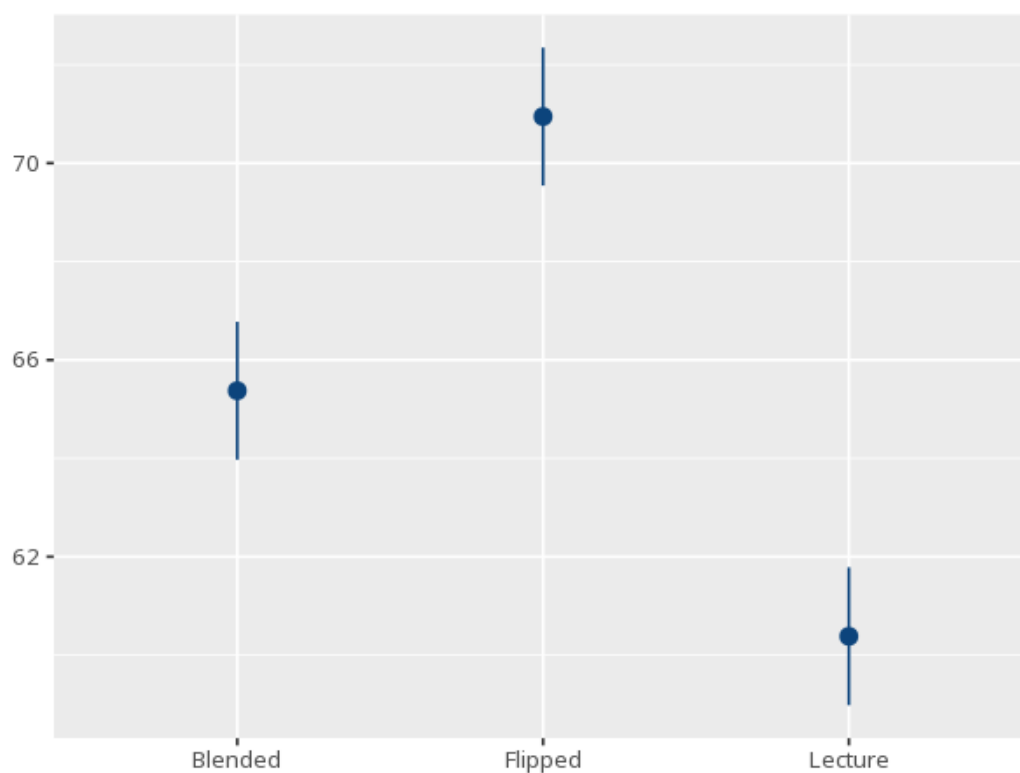


1 ANCOVA

Group	Adj. M	SE	95% CI
Blended	65.37	0.71	[63.97, 66.78]
Flipped	70.95	0.71	[69.54, 72.35]
Lecture	60.38	0.71	[58.98, 61.78]

Source	SS	df	MS	F	p	partial η^2 [95% CI]
pretest_score	2086.72	1	2086.72	51.49	<.001	0.18 [0.11, 1.00]
teaching_method	4463.20	2	2231.60	55.07	<.001	0.32 [0.24, 1.00]

Pair	Mdiff (adj.)	SE	padj	95% CI
Blended - Flipped	-5.57	1.01	<.001	[-7.95, -3.20]
Blended - Lecture	4.99	1.01	<.001	[2.62, 7.37]
Flipped - Lecture	10.57	1.01	<.001	[8.19, 12.94]



Compares group means after statistically removing the influence of a numeric covariate. First confirm the factor×covariate interaction is non-significant (homogeneity of slopes) - if it is significant, the adjusted-means interpretation is not valid and a different model is needed. Then read the Factor row's F/p for the adjusted group effect, and the adjusted means for the covariate-controlled

group values. Your significance threshold (α) is 0.05.

Controlling for the covariate, an ANCOVA gave $F(2,236)=55.07$, $p < .001$, partial $\eta^2=.32$ [0.24, 1.00].

Statistical basis: ANCOVA (group means adjusted for a covariate) (Fisher, R. A. (1925). *Statistical Methods for Research Workers*. Oliver & Boyd.); Type III sums of squares (car::Anova) (Fox J, Weisberg S (2019). *An R Companion to Applied Regression*, Third edition. Sage, Thousand Oaks CA.); Estimated marginal (covariate-adjusted) means (Lenth RV (2024). *emmeans: Estimated Marginal Means, aka Least-Squares Means*. R package.); Partial η^2 effect-size benchmarks (Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences* (2nd ed.). Routledge.).